Object Oriented Programming and Design Methodologies

CS/CE 224/272 L1
Fall Semester 2024

Hardware Prerequisites (if any): Computer with mic and camera.

Software Prerequisites (if any): C++ Compiler (g++), SDL library, VSCode as IDE

Content Area: This course meets requirements for CS Major, EE Major, CE Major

Instructor Information

Instructor: Syed Hammad Ahmed
Title: Assistant Professor
Office Location: Faculty Pod W (W-224)
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Office Hours: Tuesdays 11:30 - 1:30 pm

Instructor: Syed Mujtaba Hassan Naqvi
Title: Research Assistant
Office Location: Faculty Pod C (C-100)
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Office Hours: Tue: 11:00 - 12:00 | Thurs: 11:00 - 12:00 | Fri: 10:00-11:00

Instructor: Laiba Jamil
Email: laiba.jamil@sse.habib.edu.pk

Course Description

Object Oriented Programming (OOP) offers a new and powerful way to cope with complexity. Instead of viewing a program as a series of steps to be carried out, it views it as a group of objects that have certain properties and can take certain action. This course aims to introduce fundamental OOP concepts, and use C++ as an object oriented programming language. Students will design and develop a project applying all the concepts learned in the course. On successful completion of the course, the students are expected to have learned how complex projects are designed and developed using OOP approach.

Course Aims

OOP is a way of organizing programs. The emphasis is on the way programs are designed, not on coding details. In particular, OOP programs are organized around objects, which contain both data and functions that act on that data. This course provides motivation for using C/C++, as opposed to other languages like Java, C# and Python. It covers the basic OOP concepts in details: Classes, Objects, Constructors, Destructors, Encapsulation, Abstraction, Inheritance, and Polymorphism. It motivates to use STL library and containers of C++. Finally, students use these topics in creating a working application applying the object oriented programming techniques that are learned in the course. Designing will be a major part of the entire activity; hence UML 2.0 is learned to design the project in the end of course.

Course Learning Outcomes (CLOs)

CLO	Description	Learning-Domain Level
CLO 1	Use C++ as an Object Oriented Programming language, and its memory management.	Cog – 3
CLO 2	Investigate the concept of Object Oriented Programming with extensive programming practice	Cog – 4
CLO 3	Apply Software Design and Design Patterns by Using UML 2.0	Cog – 3
CLO 4	Create a modular project using OOP techniques	Cog – 6

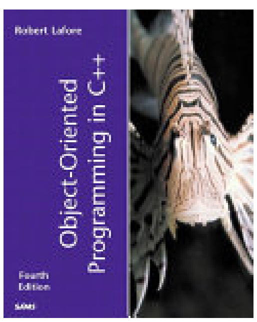
Mode of Instruction

The classes will be held on campus. The LMS (Canvas) site will be used to share the syllabus, give out assignments, and to share other course resources. Students have to read prescribed sections from the textbook before coming to class.

Engagement & Participation Rules

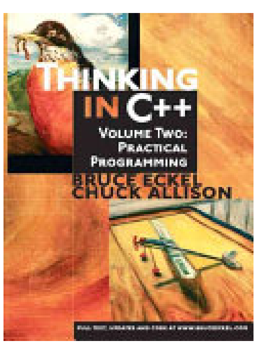
1. Official course communication will take place over the course discussion forum (LMS and MS Teams) and email. It is your responsibility to stay up to date with it. Please use the course forum to discuss all course related matters and queries, including ambiguities in assignment questions. Please respect deadlines. Submit your work by the indicated time. Incomplete work will receive partial credit.
2. You are expected to maintain a behavior befitting Yohsin and acknowledging the classroom as a place of learning, exploration, and experimentation. The University's standard policies on attendance, inclusivity, office hours, and academic integrity apply in this course.
3. Students are required to attend classes and submit assignments on time. Before each class, they should review the topics covered in the last class. Students' performance will be assessed using various instrument including exams, assignments, project, and class participation.
4. Assignments on different topics will be given during the semester. There will a project towards end of the course in which students will apply the learned techniques to solve some real-world problem.

Required Texts and Materials



Object-Oriented Programming in C++
ISBN: 9780132714297
Authors: Robert Lafore
Publisher: Pearson Education
Publication Date: 1997-12-18

Optional Materials



Thinking in C++
ISBN: 9780130353139
Authors: Bruce Eckel, Chuck D. Allison
Publisher: Prentice Hall
Publication Date: 2003-12-01

XCode and SDL 2.0 will be used (You can also use vscode with MinGW, Dev C++ or Visual Studio at your home PCs if you like).

Assessments

Object Oriented Programming & Design Methodologies course is composed of intense programing exercises and assignments. Some of the assignments, and final project will be given as a group activity. RAs can call anyone in the group for viva. It is strongly recommended that you start working on the assignment the very day you get it. Assignments WILL take time, so the sooner you start the better. Following assessments are used in the course:

Assessments	Weightage
Class Quizzes [4 quizzes]	10%
Class Assignments [3 Assignments]	15%
Midterm Exam	20%
Final Exam	20%
Lab Assignments	10%
Lab Quizzes	15%
Project	10%

Grading Scale

Letter Grade	GPA Points	Percentage
A+	4.00	[95-100]
A	4.00	[90-95)
A-	3.67	[85-90)
B+	3.33	[80-85)
B	3.00	[75-80)
B-	2.67	[70-75)
C+	2.33	[67-70)
C	2.00	[63-67)
C-	1.67	[60-63)
F	0.00	[0, 60)

Note: [a, b) is a range of numbers from a to b where a is included in the range and b is not.

Late Submission Policy

Submit your work by the indicated time. Incomplete work will receive partial credit. Late work will not be accepted or graded. Concerns regarding a score will be entertained by the respective instructor up to a week after the release of the score. Concerns raised later will not be entertained. Requests for grace marks for whatever reason will not be entertained and each such request will result in a penalty of 1% from the overall score.

Week-Wise Schedule (Tentative)

Fall 2024 Weekly Schedule*

Week	Start Date	End Date	Holidays	Topics	Readings	Assessment	Lab
1	19 Aug 2024	23 Aug 2024		Lec 1: Syllabus discussion , Introduction to OOP Lec 2: hello world program, c++ types, basic function, difference between definition and declaration. compiler, linker anatomy, loops, conditions/control statements/logical operators	Ch. 1, 2 , 3		Lab1 Introduction to the course, C++ syntax, types, size, memory Loops, Conditions, switch-case, random numbers, control statements, logical operators
2	26 Aug 2024	30 Aug 2024	26 Aug 2024 (Chehlum)	Lec 01 : Arrays and Pointers Lec 02 : Dynamic Memory allocation (using new and delete operators)	Ch. 7 , 10	HW 1 released (end of the week)	Lab 2: Arrays, Passing arrays in functions
3	2	6 Sep		Lec 01: Functions(pass	Ch. 5	Class Quiz	Lab3 -

	Sep 2024	2024		by value/references). Difference between references and pointers. Debugging - step through code to figure out what has gone wrong Lec 02 : Function Overloading		01	Pointers and Dynamic memory Lab Quiz01 : Arrays and Pointers
4	9 Sep 2024	14 Sep 2024		Lec 01 : Introduction to Classes and Objects Lec 02: public/private members functions and variable.	Ch 6	HW 1 Due	Lab4: Function overloading and pass by value/ref Const function
5	16 Sep 2024	20 Sep 2024	16 Sep 2024 (12 Rabiul Awwal)	Lec 01 : ctor/dtors , ctor overloading Lec 02 : RAII , rule of 5	Ch. 6 , for RAii : instructors will provide the material	HW 2 released (end of the week)	Lab Quiz02: Functions , Pointers , dynamic memory allocation, vectors + Review
6	23 Sep 2024	27 Sep 2024		Lec 01 : LinkedList, Stacks, Queues Lec 02 : Operator overloading	Ch. 8, for LinkedList,stack , queues : instructors will provide the material	Class Quiz 02	Lab5 (Classes and objects)
7	30 Sep 2024	4 Oct 2024		Lec 01: Inheritance , types of inheritance Lec 02 : Continue inheritance	Ch. 9	HW 2 due	Lab6 : Constructors and destructors , Constructor overloading
8	7 Oct	11	12 Oct	Lec 01: Virtual	Ch. 11	Midterm	Lab7

	2024	Oct 2024	2024 (No Classes)	Functions Lec 02 : Polymorphsim (override functions)		Exam	
9	14 Oct 2024	18 Oct 2024	14 Oct 2024 and 15 Oct 2024 - No classes - Mid semester recess	Lec 01 : Static members , Design patterns Lec 02 : Continue Design patterns	Ch. 6 , for design patterns instructors will provide the material	HW 3 released	Lab Quiz03 : pointer, dynamic array creation and deletion, operator overloading, classes, constructors, + Review
10	21 Oct 2024	26 Oct 2024		Templates in C++	Ch. 14	Project Proposal Submission Due	Lab 8
11	4 Nov 2024	8 Nov 2024	9 Nov 2024 (Iqbal Day)	Friend Functions and friend classes	Ch. 11	HW 3 Due	Lab 9
12	11 Nov 2024	15 Nov 2024		Introduction to Relationships between Objects (association, composition, aggregation)	instructors will provide the material		Lab10
13	18 Nov 2024	22 Nov 2024		Implementation of Object relationships, Class diagrams	instructors will provide the material	Project Progress - 1	Lab Quiz04 : polymorphism, inheritance, operator overloading + Review
14	11 Nov 2024	15 Nov 2024		Activity on Class diagrams (real world scenarios)	instructors will provide the material		Lab 11
15	25 Nov 2024	29 Nov 2024		Advanced topic in C++ (instructor specific)	instructors will provide the material	Project Progress - 2	Lab 12
16	30 Nov	3 Dec 2024		Project Demos			

	2024						

Notes:

* The University reserves the right to correct typographical errors or to adjust the Academic Calendar at any time it deems necessary.

† Subject to the sighting of the new moon.

‡ No Class(es).

Attendance Policy

Students are expected to read the recommended book sections/chapters and attend all online synchronous / in-person sessions. It is mandatory to attend synchronous class in-person (or online if it's determined by the university due to prevailing situation). You can be dropped from the course if attendance falls below 85% percent in any of the course component: Lecture or Lab.

Final Exam Policy

TBD

Academic Integrity

Each student in this course is expected to abide by the Habib University Student Honor Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work.

Scholastic dishonesty shall be considered a serious violation of these rules and regulations and is subject to strict disciplinary action as prescribed by Habib University regulations and policies. Scholastic dishonesty includes, but is not limited to, cheating on exams, plagiarism on assignments, and collusion.

a. Plagiarism: Plagiarism is the act of taking the work created by another person or entity and presenting it as one's own for the purpose of personal gain or of obtaining academic credit. As per University policy, plagiarism includes the submission of or incorporation of the work of others without acknowledging its provenance or giving due credit according to established academic practices. This includes the submission of material that has been appropriated,// bought, received as a gift, downloaded, or obtained by any other means. Students must not, unless they have been granted permission from all faculty members concerned, submit the same assignment or project for academic credit for different courses.

b. Cheating: The term cheating shall refer to the use of or obtaining of unauthorized information in order to obtain personal benefit or academic credit.
c. Collusion: Collusion is the act of providing unauthorized assistance to one or more person or of not taking the appropriate precautions against doing so.

All violations of academic integrity will also be immediately reported to the Student Conduct Office.

You are encouraged to study together and to discuss information and concepts covered in lecture and the sections with other students. You can give "consulting" help to or receive "consulting" help from such students. However, this permissible cooperation should never involve one student having possession of a copy of all or part of work done by someone else, in the form of an e-mail, an e-mail attachment file, a diskette, or a hard copy.

Should copying occur, the student who copied work from another student and the student who gave material to be copied will both be in violation of the Student Code of Conduct.

If you wish to use generative-AI tools to complete any of your assessments, you must first obtain permission from your course instructor. AI generated work will not be accepted in all classes or even all assessments. The instructor's permission is required. If the permission is granted, you should declare its use and properly cite the source of the generated content. Failing to identify AI written or assisted work is academic dishonesty and will be treated as any case of plagiarism by the university.

The principle for academic integrity is that your submissions must be substantially your own work and that any work that is not originally your thought must be identified and credited. If the use of AI tools is prohibited in the course, respect the rules and do not use these tools for assessments. The fundamental purpose of assessment is to learn, synthesize information and explain new connections and interpretations that arise from your secondary research. Be aware that unauthorized use of AI tools for assessments can result in a conduct case being filed. This can have serious consequences for your academic standing and future career opportunities.

During examinations, you must do your own work. Talking or discussion is not permitted during the examinations, nor may you compare papers, copy from others, or collaborate in any way. Any collaborative behavior during the examinations will result in failure of the exam, and may lead to failure of the course and University disciplinary action.

Penalty for violation of this Code can also be extended to include failure of the course and University disciplinary action.

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Program Learning Outcomes (For Administrative Review)

Upon graduation, students will have the following abilities:

- PLO 1: Theoretical Computer Science: recall and apply foundational principles of computer science.

- PLO 2: : Application Development: build software systems of varying complexity in light of fundamental computer science principles and any other constraints.

- PLO 3: Analysis and Design: perform technical analysis and design using core computing and mathematical knowledge.

- PLO 4: Systems: apply the knowledge of computing systems.

- PLO 5: : Research and Exploration: develop expertise in and contribute to a given sub-field of computing by drawing upon a strong foundation in the fundamentals of computer science and mathematics to solve real-life problems.

- PLO 6: Problem Solving: identify and analyze problems and propose effective computing-based solutions.

- PLO 7: Practical Exposure: make effective use of current tools, technologies, and good industry practices.

- PLO 8: Responsible Citizenship: conduct their computing practice in a manner that is ethical and socially responsible and corresponds to their distinct sense of identity and service to the community.

- PLO 9: Self-Learning: continuously adapt their skills to the changes taking place around them.

- PLO 10: Design Thinking: apply design thinking principles to the design of a solution.

- PLO 11: Multi-disciplinarity: incorporate knowledge and input from multiple disciplines.

- PLO 12: Communication and Teamwork: communicate and function effectively as a member or a leader of a variety of teams.

Program Learning Outcomes (PLOs) mapped to Course Learning Outcomes (CLOs)					
	CLOs of the course are designed to cater following PLOs: **PLO 1:** **PLO 2:**				
	Distribution of CLO weightages for each PLO				
	CLO 1	**CLO 2**	**CLO 3**	**CLO 4**	
PLO 2			50%	50%	
PLO 3	50%	50%			
PLO 4		100%			
PLO 8				100%	

Mapping of Assessments to CLOs

Assignments	**CLO #01**	**CLO #02**	**CLO #03**	**CLO #04**	
HW1	X				
HW2		X			
HW3			X		
Midterm Exam	X				
Final Exam		X			
Lab Quiz 1	X				
Lab Quiz 2	X				
Lab Quiz 3		X			
Lab Quiz 4		X			
Project Design			X		
Project Final Submission				X	

Recording Policy

Only asynchronous and synchronous online sessions will be conducted and recorded via MS Teams. Link to the recordings will be available to all students on Canvas Learning Management System.

Accommodations for Students with Disabilities

In compliance with the Habib University policy and equal access laws, I am available to discuss appropriate academic accommodations that may be required for student with disabilities. Requests for academic accommodations are to be made during the first two weeks of the semester, except for unusual circumstances, so arrangements can be made. Students are encouraged to register with the Office of Academic Performance to verify their eligibility for appropriate accommodations.

Inclusivity Statement

We understand that our members represent a rich variety of backgrounds and perspectives. Habib University is committed to providing an atmosphere for learning that respects diversity. While working together to build this community we ask all members to:

- share their unique experiences, values and beliefs
- be open to the views of others
- honor the uniqueness of their colleagues
- appreciate the opportunity that we have to learn from each other in this community
- value each other's opinions and communicate in a respectful manner
- keep confidential discussions that the community has of a personal (or professional) nature
- use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the Habib community

Office Hours Policy

Every student enrolled in this course must meet individually with the course instructor during course office hours at least once during the semester. The first meeting should happen within the first five weeks of the semester but must occur before midterms. Any student who does not meet with the instructor may face a grade reduction or other penalties at the discretion of the instructor and will have an academic hold placed by the Registrar's Office.